

DORA'S DENTAL GEM

Dental Clinic Management System

System Blueprint & Requirements Document

Includes Automated SMS & Email Notification Module

Detail	Information
Client	Dora's Dental Gem — Level 2 K, Kabuusu, SM Mall, Kampala
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Prepared for client review and feedback prior to full-scale development.

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1. Introduction

Dora's Dental Gem currently relies on manual and physical record-keeping for patient files, appointments, billing, and inventory. This creates risks of lost records, double-booking, delayed follow-ups, and difficulty tracking clinic performance.

This document presents the blueprint for a Dental Clinic Management System (DCMS) — a digital platform that centralizes all clinic operations into one secure system. It is intended to guide the development of a working demo for client feedback, while identifying the full set of features the complete system should eventually support.

1.1 Purpose of the Demo

The demo is a working, scaled-down version of the full system. It will demonstrate the core workflows a dentist, receptionist, and admin would use daily, and gather client feedback before full-scale development begins. Sections in this document marked "Demo Phase" indicate features included in the first build; those marked "Full System Phase" are planned for later rollout.

2. Project Objectives

- Eliminate manual/paper-based patient and treatment records.
- Provide a central, searchable database of patients, appointments, treatments, and payments.
- Reduce missed appointments through automated SMS/Email reminders.
- Improve communication with patients (confirmations, updates, holiday greetings, promotions).
- Give clinic management visibility into revenue, inventory, and staff activity through reports.
- Protect patient data with role-based access and secure storage.

3. System Modules — Overview

The system is organized into the following functional modules. Each is detailed in Section 4.

#	Module	Phase
1	Patient Management	Demo Phase
2	Appointment Scheduling	Demo Phase
3	Clinical Records & Treatment Management	Demo Phase
4	Billing & Payments	Demo Phase
5	Notifications — SMS & Email	Demo Phase (core), Full System (advanced)
6	Inventory Management	Full System Phase
7	Staff Management	Demo Phase (basic), Full System (advanced)
8	Reports & Analytics	Demo Phase (basic), Full System (advanced)
9	Insurance Management	Full System Phase
10	Document & Records Storage	Demo Phase (basic), Full System (full)
11	Admin & Security	Demo Phase

4. Detailed Module Breakdown

4.1 Patient Management

Central digital record for every patient, replacing paper patient cards.

- Register new patients: names, contact, date of birth, address, next of kin.
- Store medical history: allergies, existing conditions, medications.
- Store dental history and past treatment summary.
- Search and filter patients by name, phone number, or ID.
- Edit/update patient profile over time.

4.2 Appointment Scheduling

Digital calendar replacing the appointment book.

- Book, reschedule, and cancel appointments.
- View dentist/chair availability to avoid double-booking.
- Daily/weekly appointment view for reception.
- Automatic trigger to Notifications module for reminders and confirmations.

4.3 Clinical Records & Treatment Management

Captures what happens during each visit.

- Tooth-by-tooth dental chart (odontogram).
- Diagnosis notes and treatment plan per visit.
- Treatment history log — full timeline per patient.
- Prescriptions issued.

4.4 Billing & Payments

Tracks money owed and paid, replacing manual receipt books.

- Generate invoices per treatment/visit.
- Record payments: cash, mobile money, bank, card.
- Track outstanding balances per patient.
- Generate/print receipts.

4.5 Notifications Module — SMS & Email (Client-Requested Feature)

A dedicated communication engine that keeps patients informed automatically, reducing no-shows and improving the clinic's relationship with its clients.

Trigger-based automatic messages

- Appointment reminders — sent a set time before a booked visit (e.g. 24 hrs / 2 hrs before).
- Appointment confirmation — sent immediately after a booking is made.
- Rescheduled / cancelled appointment alerts.
- Treatment/visit follow-up messages (e.g. post-extraction care reminders).
- Payment confirmation and balance-due reminders.

Broadcast / bulk messages

- Holiday and festive-season greetings to the full patient list.
- Clinic announcements — new services, working-hour changes, promotions.
- Targeted reminders, e.g. "6-month checkup due" sent to patients whose last visit was 6 months ago.

Admin controls

- Message templates that staff can edit without touching code.
- Choice of channel per message: SMS, Email, or both.
- Delivery log — record of what was sent, to whom, and when.
- Opt-out handling so patients can stop promotional (non-critical) messages.

Note: for the demo, this module will simulate sending (logging the message in-system) or use a low-cost SMS gateway sandbox and a real email service, since live SMS gateways in Uganda (e.g. Africa's Talking, EgoSMS) typically require a paid account and sender ID approval before go-live.

4.6 Inventory Management (Full System Phase)

- Track dental supplies and materials in stock.
- Low-stock alerts.
- Supplier records and reorder tracking.

4.7 Staff Management

- Staff profiles: dentists, assistants, receptionist, admin.
- Role assignment (see Section 5).
- Duty roster / shift tracking (Full System Phase).

4.8 Reports & Analytics

- Daily / monthly patient visit counts.
- Revenue summaries.
- Most common treatments/procedures (Full System Phase).
- Notification delivery reports — sent vs failed (Full System Phase).

4.9 Insurance Management (Full System Phase)

- Record patient insurance provider and policy details.
- Track and log insurance claims.

4.10 Document & Records Storage

- Upload and attach X-rays, scanned forms, and consent documents to a patient's file.
- Secure, centralized storage — replacing physical filing cabinets.
- Automatic backup (Full System Phase).

4.11 Admin & Security

- User login with role-based access (see Section 5).
- Audit trail — logs who created/edited/deleted a record and when.
- Password protection and session control.

5. User Roles & Permissions

Role	Access
Admin	Full access — manage users, view all reports, configure notification templates.
Dentist	View/edit assigned patients, clinical records, treatment plans, prescriptions.
Receptionist	Register patients, manage appointments, trigger notifications, record payments.
Patient (future portal)	View own appointments and receive notifications only (Full System Phase).

6. Proposed System Architecture

6.1 High-Level Structure

- Frontend — web-based interface for reception/admin (and optionally a lightweight mobile view for dentists).
- Backend/API — handles business logic: appointments, records, billing, notification triggers.
- Database — stores all clinic data (patients, appointments, treatments, payments, inventory).
- Notification Service — integrates with an SMS gateway (e.g. Africa's Talking / EgoSMS) and an email service (e.g. SMTP/SendGrid) to dispatch messages triggered by system events.

6.2 Suggested Technology Stack (Demo)

Layer	Suggested Technology
Frontend	React / HTML-CSS-JS web app
Backend	Node.js (Express) or PHP
Database	MySQL / PostgreSQL
SMS Gateway	Africa's Talking or EgoSMS (Uganda-friendly)
Email Service	SMTP (Gmail/SendGrid) for automated emails
Hosting (demo)	Local server / low-cost cloud host for presentation

Final stack to be confirmed based on client hosting budget and preference.

7. Core Database Entities (Preview)

A full Entity-Relationship Diagram (ERD) will be produced during the design phase. Core entities include:

- Patients
- Appointments
- Treatments / Dental Chart Records
- Invoices & Payments
- Staff / Users & Roles
- Notification Log (message, channel, recipient, status, timestamp)
- Inventory Items (Full System Phase)

8. Non-Functional Requirements

Requirement	Description
Security	Role-based login; patient data access restricted by role; passwords hashed.
Reliability	Notification failures logged and retried; no silent message loss.
Usability	Simple interface for non-technical reception staff.
Data Integrity	No permanent deletion of clinical records — soft-delete with audit trail.
Scalability	System should handle growth from one branch to multiple branches later.
Availability	Demo runs locally/on a test server; full system targets near 24/7 uptime.

9. Demo Scope Summary

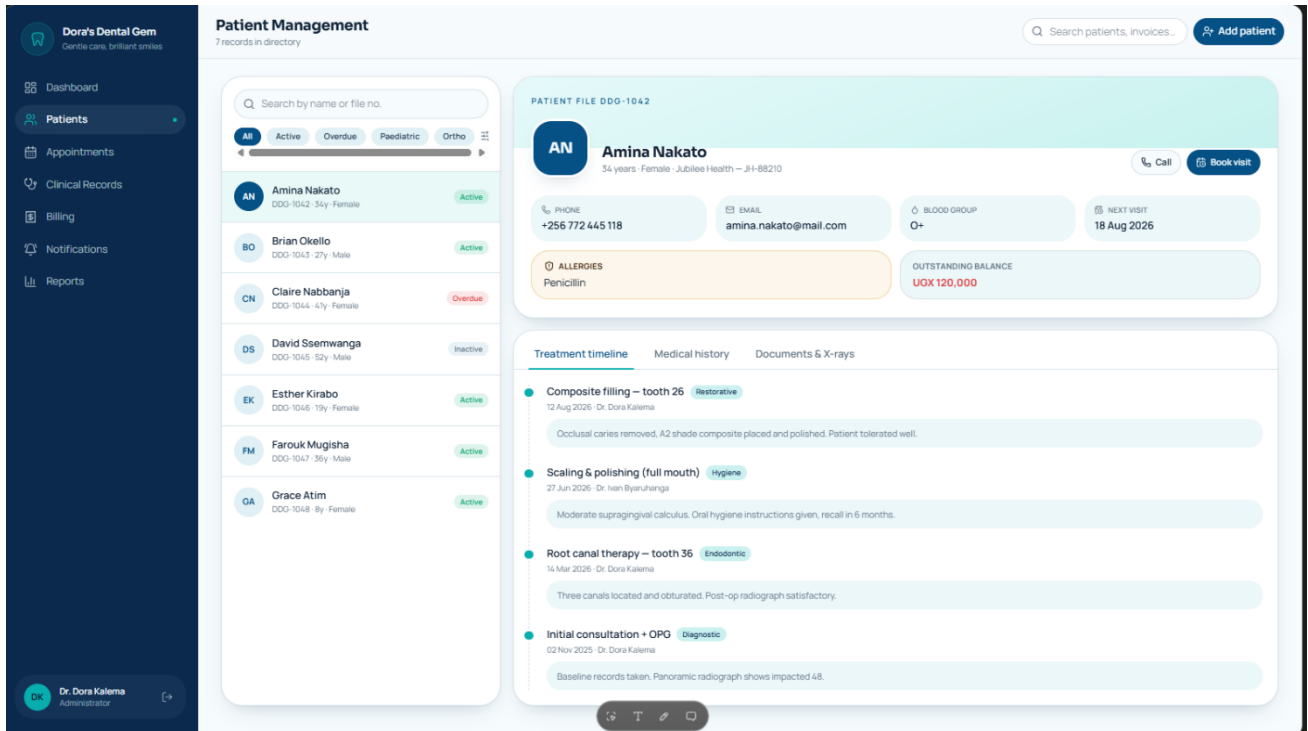
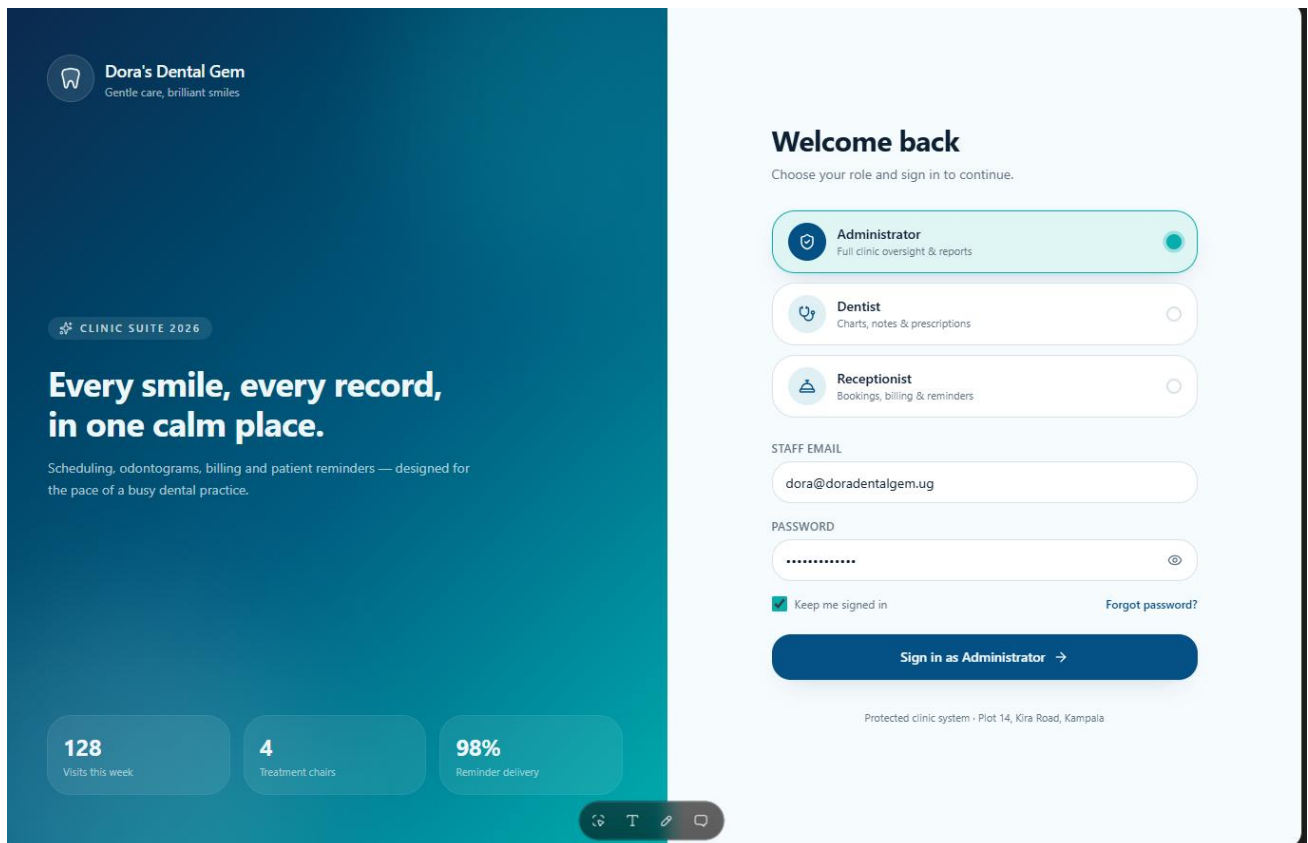
To keep the presentation focused, the demo build will prioritize the workflows a client can see and interact with end-to-end:

- Register a patient and view their profile.
- Book an appointment and see a confirmation notification triggered (SMS/Email).
- Record a basic treatment note against a patient.
- Generate a simple invoice and mark it paid.
- Send a sample bulk notification (e.g. holiday greeting) to a test patient list.
- Admin login showing role-based access.

Modules marked "Full System Phase" throughout this document (Inventory, Insurance, advanced Reports, Patient Portal) are excluded from the demo but retained in this blueprint to guide the complete build after client feedback.

10. Next Steps

- Client review of this blueprint and feedback on priorities.
- Confirm technology stack and hosting preference.
- Confirm preferred SMS gateway/provider and email sender details.
- Begin database design (ERD) and UI wireframes for the demo.
- Build and present working demo.



**Dora's Dental Gem**
Gentle care, brilliant smiles

Dashboard

Patients

Appointments

Clinical Records

Billing

Notifications

Reports

OK

Dr. Dora Kalema

Administrator

Appointment Scheduling

Tuesday, 18 August 2026 - Day view

Q Search patients, invoices...

+ Book

SAT 15
7

SUN 16
0

MON 17
11

**TUE 18
9**

WED 19
6

THU 20
5

FRI 21
10

Confirmed Pending Completed Cancelled

TIME	Chair 1 Dr. Dora	Chair 2 Dr. Ivan	Chair 3 Dr. Ivan
08:00	Amina Nakato Review + polish 08:00 - 30 min - Dr. Dora		
08:30		Brian Okello Night guard fitting 08:30 - 60 min - Dr. Ivan	
09:00			
09:30	Esther Kirabo Ortho adjustment 09:30 - 30 min - Dr. Dora		
10:00		Farouk Mugisha Deep scaling 10:00 - 60 min - Dr. Ivan	
10:30			
11:00	Grace Atim Fluoride sealants 11:00 - 30 min - Dr. Dora		
11:30			
12:00			Claire Nabbanja Crown prep 12:00 - 30 min - Dr. Dora

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OK

Dr. Dora Kalema

Administrator

Clinical Records

Amina Nakato - DDG-1042 - Session 12 Aug 2026

Q Search patients, invoices...

Print chart

Odontogram - FDI notation

Tap any tooth to chart a finding

Permanent dentition

18 17 16 15 14 13 12 11

21 22 23 24 25 26 27 28

48 47 46 45 44 43 42 41

51 52 53 54 55 56 57 58

Healthy Filled Caries Crown Root canal Missing

**Tooth 26**

Current status: Filled - Surfaces: O, M

+ Caries

+ Filling

+ Crown

+ Root canal

+ Extraction

+ Sealant

Treatment notes

12 Aug 2026 - 09:14 - Dr. Dora Kalema

Chief complaint: sensitivity upper left on cold. Examination reveals occlusal caries on 26 and early demineralisation on 24. Vitality tests within normal limits.

12 Aug 2026 - 09:52 - Dr. Dora Kalema

Caries excavated under rubber dam on 26. Class I composite placed (A2). Occlusion checked and adjusted. Patient advised on fluoride varnish for 24 at next recall.

Add a clinical note for tooth 26...

Save note

Prescriptions

+ Add

Amoxicillin 500mg

1 capsule TDS - 5 days

After meals

Metronidazole 400mg

1 tablet TDS - 5 days

Avoid alcohol

Chlorhexidine 0.2% rinse

10ml BD - 7 days

Do not swallow

Print prescription slip

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Dashboard

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Dr. Dora Kalema

Administrator

Billing & Payments
Invoice INV-2026-0189 - draft

UGX 24,300,000
Collected this month

UGX 4,820,000
Outstanding balance

UGX 1,940,000
Pending insurance

Dora's Dental Gem
Plot 14, Kira Road, Kampala

INVOICE #2026-0189

BILLED TO	ISSUED	DUE	
Amina Nakato · DDG-1042	18 Aug 2026	01 Sep 2026	

CODE	DESCRIPTION	QTY	AMOUNT
D2391	Composite filling – 1 surface (tooth 26)	1	180,000
D1118	Adult prophylaxis / scaling	1	120,000
D8274	Bitewing radiographs (x2)	2	70,000
D9238	Local anaesthesia	1	25,000
Subtotal			UGX 395,000
VAT (18%)			UGX 71,100
Insurance cover			- UGX 150,000
Total due			UGX 316,100

Download PDF

Email invoice

Record payment

Payment history

Last 30 days

Amina Nakato	INV-2026-0189 · 12 Aug 2026 · Mobile Money	395,000	Paid
Farouk Mugisha	INV-2026-0187 · 11 Aug 2026 · Card	80,000	Partial
Claire Nabbanja	INV-2026-0185 · 09 Aug 2026 · Insurance	340,000	Overdue
Brian Okello	INV-2026-0185 · 05 Aug 2026 · Cash	220,000	Paid
Esther Kirabo	INV-2026-0184 · 03 Aug 2026 · Mobile Money	55,000	Partial
Grace Atim	INV-2026-0183 · 01 Aug 2026 · Cash	140,000	Paid

Ageing of receivables

0-30 days	62%
31-60 days	24%
61-90 days	9%
90+ days	5%

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Dr. Dora Kalema

Administrator

Notifications Center
1,284 patients reachable · 98% delivery rate

TEMPLATES

Appointment reminder
SMS template

Booking confirmation
SMS template

6-month recall
Email template

Holiday greeting
Email template

Outstanding balance
SMS template

+ New template

Appointment reminder
Merge tags auto-fill from the patient record

SMS

Email

RECIPIENTS

Amina Nakato · Brian Okello · Claire Nabbanja · David Ssemwanga · +38 tomorrow's list

MESSAGE

Hi Amina, this is a reminder of your appointment at Dora's Dental Gem on 18 Aug 2026 at 09:30 with Dr. Dora. Reply YES to confirm.

130 chars · 1 SMS credit

SEND WINDOW

Today · 18:00 EAT

ESTIMATED COST

UGX 1,680 (42 recipients)

Send sms to 42 patients

LIVE PREVIEW

DORAGEM · SMS

Hi Amina, this is a reminder of your appointment at Dora's Dental Gem on 18 Aug 2026 at 09:30 with Dr. Dora. Reply YES to confirm.

Delivery log

Live

Amina Nakato	SMS · Appointment reminder · Today · 07:02	Delivered
Brian Okello	SMS · Booking confirmation · Today · 06:58	Delivered
claire.n@gmail.com	Email · Outstanding balance · Today · 06:40	Opened
Esther Kirabo	SMS · Appointment reminder · Yesterday · 18:...	Pending
David Ssemwanga	SMS · 6-month recall · Yesterday · 17:55	Failed
guardian.atim@mail.com	Email · Holiday greeting · 16 Aug · 09:20	Delivered

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